Industrial Quality Control – Automated Defect Detection

Scenario

An industrial manufacturer specializing in precision components wants to automate its quality control process on the production line. Currently, manual inspection is slow and prone to human error. The goal is to deploy a computer vision system capable of identifying structural or surface defects in real-time.

As their solution provider, you are expected to design a solution answering their needs with evidence of its quality. Since you only have a day, a POC and detailed report are requested instead of the whole solutions.

Reminder : the deadline is strict, you must submit your propositions by 20h00 today, otherwise the customer will apply penalties on your pay. You must be accurate, detailed, yet fast on your production.

The company data are known as an industry-standard datasets: MVTec AD (Anomaly Detection Dataset) - containing 15 different categories of objects and textures with various defect types (total 4.9 Go) : https://www.mvtec.com/research-teaching/datasets/mvtec-ad

The more defaults your solution accurately detects, the better. Each good product evaluated to be a defect is a direct loss of money for the company, each defect evaluated to be defect-free is a reputation loss to our customers and a set of procedures. Likewise, the more kind of defects the solution can identify, the more money the company can save (and thus accept to pay you more).

Important note: When downloading the data, you may notice that an evaluation code is provided – You are not to use it, building the evaluation procedure of your solution is part of the job, using the existing code would be considered a fraud.

You have 1 day, you can do it, good luck!

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